C

ACT SUBJECT TEST — MATH 2 • UNIT 1.01

# Complex Numbers & De Moivre Theorem

PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

## Complex Numbers & De Moivre Theorem Practice Exercises

**1** MCQ

For the complex number $i = \sqrt{-1}$, which of the following is equivalent to the expression $(5 - 3i) - (2 - 7i)$?

A $3 + 4i$

B $3 - 10i$

C $7 + 4i$

D $3 - 4i$

**2** MCQ

For what real value of $k$ is the complex expression

$$\frac{k + 8i}{3 - i} + \frac{2 - ki}{3 + i}$$

a real number?

A 11

B 1

C 6

D -11

**3** MCQ

Let $z$ be a complex number satisfying the equation

$$|z - 12| + |z + 12| = 26.$$

What is the maximum possible value of $|z - 5i|$?

A $\frac{169}{12}$

B $\sqrt{194}$

C $18$

D $10$

**4** MCQ

For the imaginary unit $i$ such that $i^2 = -1$, what is the product $(4 - 3i)(2 + 5i)$?

A $23 + 14i$

B $-7 + 14i$

C $23 - 14i$

D $8 - 15i$

**5** MCQ

For the imaginary unit $i$, where $i^2 = -1$, the expression $\frac{k+6i}{3-2i}$ is equal to a real number for some real constant $k$. What is the value of $k$?

A -9

B -4

C 4

D 9

**6** MCQ

In the complex plane, the complex numbers $z$ and $w$ satisfy $|z| = 5$, $|w| = 7$, and $|z + w| = 10$. What is the value of $|z - w|$?

A $4\sqrt{3}$

B $2\sqrt{6}$

C $4\sqrt{6}$

D $2\sqrt{37}$

**7** MCQ

For the imaginary unit $i$, where $i^2 = -1$, which of the following complex numbers is equivalent to $(4 + 3i)(2 - 5i)$?

A $23 - 14i$

B $23 + 14i$

C $-7 - 14i$

D $-7 + 14i$

**8** MCQ

In the complex number system, let $i^2 = -1$. For what real value of $k$ is the expression

$$\frac{6 + ki}{3 + 2i}$$

a purely imaginary number?

A -9

B -4

C 4

D 9

**9** MCQ

In the complex plane, let $z$ be a complex number that satisfies the equation

$$|z - 4| + |z + 4| = 10.$$

What is the maximum possible value of $|z - 3i|$?

A $\sqrt{34}$

B $\frac{13}{2}$

C 6

D $\frac{25}{4}$

**10** MCQ

For the imaginary unit $i$, where $i^2 = -1$, which of the following is equivalent to the expression $(5 - 2i)(3 + 4i)$ ?

A $7 + 14i$

B $23 + 26i$

C $23 + 14i$

D $7 - 14i$

**11** MCQ

What is $i^4$?

A $i$

B $-i$

C $-1$

D $1$

**12** MCQ

Simplify $(2 + 3i) + (5 - i)$.

A $7 + 2i$

B $-3 + 2i$

C $7 - 2i$

D $7 + 4i$

**13** MCQ

Simplify $(4 - 2i) - (1 + 3i)$.

A $3 + i$

B $5 - 5i$

C $3 - 5i$

D $3 + 5i$

**14** MCQ

What is the imaginary part of $z = 6 - 7i$?

A $6$

B $-7$

C $7$

D $-6$

**15** MCQ

Simplify $i \cdot i \cdot i$.

A $i$

B $1$

C $-i$

D $-1$

**16** MCQ

Simplify $3i \cdot 4i$.

A $12$

B $12i$

C $-12$

D $-12i$

**17** MCQ

Find $|3 + 4i|$.

A $5$

B $7$

C $25$

D $\sqrt{7}$

**18** MCQ

What is the complex conjugate of $5 - 2i$?

A $5 - 2i$

B $-5 - 2i$

C $5 + 2i$

D $-5 + 2i$

**19** MCQ

Multiply $(2 + i)(3 - i)$.

A $7 - i$

B $5 + i$

C $7 + i$

D $6 + i$

**20** MCQ

Simplify $\dfrac{1 + i}{1 - i}$.

A $-i$

B $1$

C $i$

D $-1$

**21** MCQ

Simplify $i^7$.

A $i$

B $-i$

C $1$

D $-1$

**22** MCQ

Find $|z|$ for $z = 5 - 12i$.

A $\sqrt{17}$

B $17$

C $7$

D $13$

**23** MCQ

If $(x + 2) + (y - 3)i = 5 + 4i$ for real $x, y$, find $x + y$.

A $9$

B $10$

C $8$

D $12$

**24** MCQ

Simplify $(2 - 3i)^2$.

A $-5 + 12i$

B $13 - 12i$

C $-5 - 12i$

D $5 - 12i$

**25** MCQ

If $z = 4 + 3i$, find $z\bar{z}$.

A $5$

B $-7$

C $25$

D $7$